Phishing Email Detection & Awareness Report

Cyber Security Internship – Task 2

📄 introduction

Phishing is a social engineering attack in which attackers impersonate trusted organizations to trick users into revealing sensitive information such as passwords, OTPs, or banking details. These attacks typically use urgency, fear, or reward-based manipulation.

This report analyzes phishing email samples to identify indicators, classify risk, and provide awareness guidelines.

📄 Email Analysis & Phishing Indicators

📧 Email Sample 1 – Account Suspension Alert

✓ Identified Phishing Indicators:

- Urgent subject line ("Account Will Be Locked")
- Fear-based message
- Generic greeting ("Dear User")
- Suspicious link (secure-account-verify.com)
- HTTP (not secure HTTPS)

Risk Classification: Phishing

How the Attack Works:

The attacker creates urgency and fear to force the user to click a fake link. The link leads to a fraudulent website designed to steal login credentials.

📧 Email Sample 2 – Bank Transaction Alert

✓ Identified Phishing Indicators:

- Financial scare tactic
- Short response deadline (12 hours)
- Suspicious domain
- Immediate action request

Risk Classification: Phishing

How the Attack Works:

The email attempts to scare the user by claiming an unauthorized transaction. Clicking the link may redirect to a fake banking portal to capture credentials.

📧 Email Sample 3 – Instagram Security Alert

✓ Identified Phishing Indicators:

- Password reset threat
- Suspicious login link
- Domain mismatch
- Account suspension warning

Risk Classification: Phishing

How the Attack Works:

The attacker impersonates Instagram and tricks the user into entering login details on a fake website.

📧 Email Sample 4 – Lottery Reward Scam

✓ Identified Phishing Indicators:

- Unrealistic reward
- Time pressure (24 hours)
- Request for bank details
- Suspicious website link

Risk Classification: Phishing

How the Attack Works:

The attacker exploits greed by offering a fake reward and requests financial information.

📧 Email Sample 5 – Fake Job Offer

✓ Identified Phishing Indicators:

- Requests Aadhaar and PAN details
- No official company domain
- Sensitive document request

⚠ **Risk Classification:** Phishing

How the Attack Works:

The attacker uses a fake job opportunity to collect personal identification documents for fraud.

📄 Prevention Tips for Users

- Always check sender email domain carefully.
- Hover over links before clicking.
- Verify suspicious emails through official websites.
- Enable multi-factor authentication.
- Report suspicious emails immediately.

📄 Do's and Don'ts for Employees

Do's:

- Verify sender identity
- Double-check URLs
- Confirm unusual requests with IT department
- Keep security awareness updated

Don'ts:

- Do not click unknown links
- Do not share OTP, passwords, or banking details
- Do not download unknown attachments
- Do not panic due to urgent language

Conclusion

Phishing attacks primarily target human behavior rather than technical vulnerabilities. By identifying phishing indicators and following proper verification practices, organizations can significantly reduce the risk of data breaches and financial fraud.